A Reference-Free Summary Evaluation System

Two Independently Implemented Plugins for Auditable Japanese News Evaluation

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AI Quality Scientist Take-Home Assignment

50 articles, 250 candidate summaries

***Abstract*—This work presents a reference-free evaluator for Japanese news summaries. The central design is a funnel: deterministic checks first identify physically verifiable contract violations; assigned-article embedding supplies mandatory relevance evidence; isolated language-model Agents then judge semantic eligibility and graded quality; and an independent Reviewer verifies every terminal decision and score. The evaluator is delivered as an executable Plugin rather than a collection of prompts. A Claude Code Plugin is the primary implementation and produces the submitted scores; a separately implemented Codex Plugin reproduces the same evaluation contract as cross-host validation. Both processed all 250 article–summary pairs. The primary run achieved 91.49% agreement with offline weak labels and zero planted-bad ranking inversions across 50 articles. The two implementations agreed on routing for 94.4% of candidates and achieved 0.8907 Spearman correlation on jointly scored candidates. These results support both the evaluation logic and its reproducibility without using reference summaries at runtime.**

***Index Terms*—LLM evaluation, summary quality, reference-free evaluation, agentic workflow, reproducibility**

I. Exploration

A. Dataset and observed populations

The dataset contains five opaque candidates for each of 50 Japanese XL-Sum articles. Candidate lengths range from 22 to 302 characters (median 125). Manual inspection and targeted probes exposed reference-quality summaries, fluent paraphrases, low-coverage outputs, changed entities and numbers, continuous source copies, unrelated text, incomplete fragments, central factual reversals, and central fabrications. A single similarity score cannot represent these populations faithfully: source copying is highly similar to the article, while on-topic reversals preserve topic vocabulary despite being semantically invalid.

B. Embedding calibration and anchor ablation

Two disjoint 10-article experiments tested `qwen3-embedding:0.6b`. Relevant pairs averaged 0.800 and 0.780; unrelated pairs averaged 0.275 and 0.263. Relevant values were 0.707–0.868 and unrelated values 0.163–0.398, so 0.50 separated all 40 probe pairs. This justified embedding as a required relevance signal, not as an autonomous verdict.

Candidate-to-anchor embedding was tested for coverage. Its small-sample Spearman correlation was higher than article similarity (0.6303 vs. 0.5282), but leave-one-out error was worse (MAE 4.0123 vs. 3.6246; combined features 4.4121). I therefore retained the structured anchor for coverage reasoning without adding a second numerical similarity feature.

TABLE I

Exploration findings translated into design decisions.

Finding	Rejected shortcut	Decision
No runtime human reference	Score against `reference_summary`	Restrict runtime evidence to article and candidate.
Copies resemble the article	Similarity as quality	Detect copying before semantic scoring.
Reversals remain on topic	Embedding as factuality	Use Agents for source-grounded semantics.
Peripheral errors remain rankable	Every factual error is terminal	Separate central invalidity from soft penalties.
Headline endings resemble cuts	Punctuation-only truncation	Terminate only an unusable fragment.

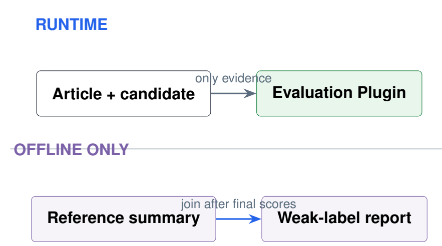


Fig. 1. Runtime and offline evidence are deliberately separated.

II. Design

A. Evidence boundary

The production evaluator never sees a human reference or another article. References join only after both runs are final, exclusively for offline validation. This prevents an evaluation shortcut that would not exist in the real application.

B. Evaluation funnel

The architecture follows four invariants: code measures and proposes while Agents judge; eligibility precedes graded quality; every consequential decision is reviewed independently; and every output retains a machine-auditable trace.

C. Physical and semantic eligibility

The deterministic gate normalizes Unicode and whitespace, counts top-level sentences outside paired delimiters, and records copy, length, and truncation evidence. Copying is proposed when the candidate is fully contained in the article

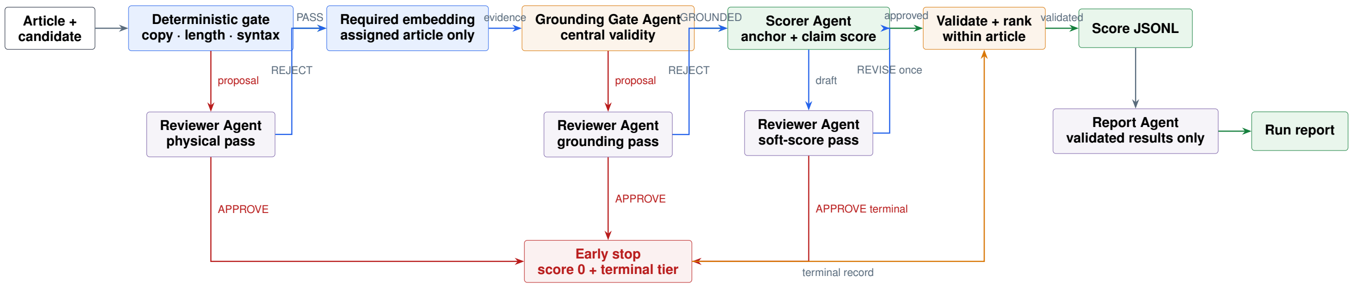


Fig. 2. Complete evaluation funnel. Each Agent role is isolated. Terminal proposals require Reviewer approval; rejection continues to the next eligible stage, while a score revision is bounded to one pass.

for at least 30 normalized characters, or when an exact span of at least 60 characters has at least 0.90 candidate coverage and 0.95 12-character-shingle coverage. More than three sentences violates the product contract. These checks propose terminal outcomes; the physical Reviewer confirms or rejects them.

Every survivor is embedded against the assigned title plus the first 1,500 article characters. One article vector is cached across its five candidates. The trace records similarity, threshold, suspect state, and zero cross-article comparisons. The Grounding Gate Agent then judges only central semantic validity. It may propose `OFF_TOPIC`, `FACTUAL_REVERSAL`, or `FABRICATED_CONTENT`, citing article and candidate spans. A Grounding Reviewer must confirm centrality before score 0 is allowed.

D. *Graded quality and independent review*

The Scorer first generates one article-only anchor—main event, key facts, and a short anchor summary—before seeing candidates. It decomposes each candidate into atomic claims and marks each `SUPPORTED`, `CONTRADICTED`, or `NOT_IN_SOURCE` with source evidence. Four dimensions sum exactly to 100: faithfulness 50, coverage 30, coherence 15, and conciseness 5. Weighting makes factual reliability dominant while preserving useful differentiation among eligible summaries.

A fresh Reviewer rechecks claim evidence, dimensions, arithmetic, label bands, and trace events. It may approve, request one bounded revision, or confirm a terminal failure missed upstream; scores are never averaged between Agents. Each Agent receives exactly one condition-matched synthetic few-shot, including a peripheral-defect counterexample that discourages over-termination.

Eligible summaries always rank above terminal outputs. Within the terminal set, unrelated, centrally false, fabricated, and empty outputs use tier 0; source copies tier 1; unusable truncations tier 2; and over-length outputs tier 3. The distinction preserves the task's need to order five candidates even when multiple candidates receive score 0.

E. *Plugin as executable specification*

The evaluator is delivered as a Plugin: one command activates the evaluation skill, deterministic scripts, isolated Agent roles, rubric, schemas, validators, and deterministic

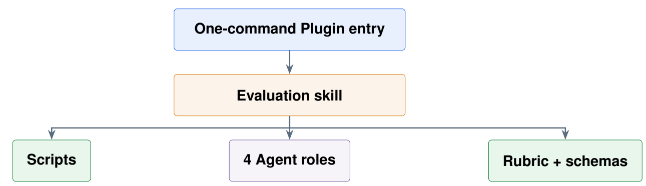


Fig. 3. The Plugin makes the evaluation design directly executable.

report generation. The Claude Code Plugin is primary and produced `scores.jsonl`. The Codex Plugin independently implements the same contract using genuine Codex subagents and extra hand-off validators. Only embedding runs locally; local generative models are prohibited for grounding, anchoring, scoring, review, revision, and report conclusions.

III. Validation

A. *Complete independent runs*

Both Plugins evaluated all 250 article–candidate pairs. All output rows passed deterministic schema, arithmetic, ordering, label-band, and trace validators. Table III reports every final outcome; Fig. 4 shows the corresponding routing and soft-label distributions.

B. *Offline weak-label validation*

References were withheld during scoring and joined only after both runs were final. This avoids using them as unavailable runtime ground truth while still exploiting their diagnostic value. Both systems retained all 58 reference reproductions; both terminated all 50 source copies and all 16 planted off-topic candidates. On 141 gradable records, the primary Claude run agreed with the offline weak labels on 129 (91.49%), and Codex agreed on 127 (90.07%). Most importantly for the five-way ranking task, neither implementation placed a planted-bad candidate above the intended survivor in any of the 50 articles.

C. *Cross-host replication*

The independent implementations provide a stronger test than rerunning one prompt. They agree on terminal versus graded routing for 236/250 candidates (94.4%). Among the 74 candidates both route to terminal, the terminal category agrees on all 74. For the 162 candidates both grade, score ordering has Spearman correlation 0.8907 and mean absolute

TABLE II

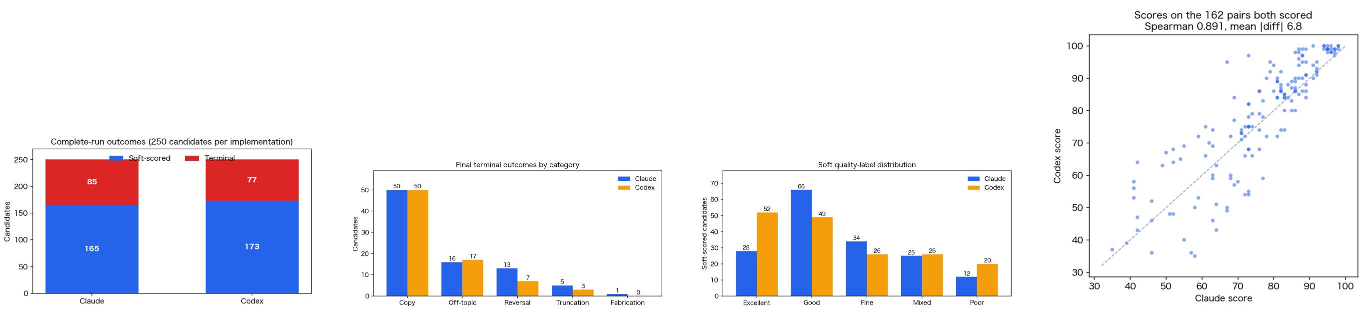
COMPLETE OUTPUT LABEL TAXONOMY. TERMINAL LABELS RECEIVE SCORE 0 BUT RETAIN A SEPARATE TIER FOR WITHIN-ARTICLE ORDERING.

Label	**Explanation**
`EXCELLENT`	Faithful, comprehensive, coherent, and concise; score 90–100.
`GOOD`	Strong and reliable with limited omissions or presentation weaknesses; score 75–89.
`FINE`	Usable and communicates the main event, with one clear gap; score 65–74.
`MIXED`	Partially useful with material factual, coverage, or readability weakness; score 50–64.
`POOR`	Eligible but substantially weak; remains usable enough to compare; score 0–49.
`HARD_FAIL_EMPTY`	No usable output.
`HARD_FAIL_OFF_TOPIC`	Unrelated to the assigned article.
`HARD_FAIL_REVERSAL`	Reverses the central event, outcome, state, or decision.
`HARD_FAIL_FABRICATION`	Central event, quotation, outcome, or load-bearing fact is unsupported.
`HARD_FAIL_COPY`	Continuous or near-complete source copy rather than a summary.
`HARD_FAIL_TRUNCATION`	Unusable fragment that stops before a complete proposition.
`HARD_FAIL_OVER_LENGTH`	Exceeds the maximum of three sentences.

TABLE III

COMPLETE-RUN OUTCOMES FOR BOTH INDEPENDENTLY ORCHESTRATED IMPLEMENTATIONS.

Run metric	**Claude**	**Codex**	**Final category**	**Claude**	**Codex**
Articles / candidates / valid rows	50 / 250 / 250	50 / 250 / 250	Verbatim source copy	50	50
Soft-scored candidates	165 (66.0%)	173 (69.2%)	Off-topic	16	17
Terminal candidates	85 (34.0%)	77 (30.8%)	Central factual reversal	13	7
Eligible mean / median	76.07 / 79	76.15 / 82	Obvious truncation	5	3
Eligible range	35–98	28–100	Central fabrication	1	0
			Over length / empty	0	0
			Terminal total	**85**	**77**



(a) Eligibility routing (b) Terminal categories (c) Eligible quality labels (d) Joint-score agreement

Fig. 4. Full-dataset validation. The implementations reproduce the same broad routing behavior and strongly preserve eligible-score ordering.

TABLE IV

OFFLINE VALIDATION AFTER FINAL SCORING.

Check	**Claude**	**Codex**
Reference reproductions survived	58/58	58/58
Source copies terminated	50/50	50/50
Off-topic candidates terminated	16/16	16/16
Weak-label agreement	129/141 91.49%	127/141 90.07%
Planted-bad inversions	**0/50**	**0/50**

difference 6.85 points. The joint-set means are also close: 76.27 for Claude and 78.23 for Codex. Thus, different Agent hosts reproduce both the funnel's routing and the relative ordering of eligible candidates.

TABLE V

AGREEMENT BETWEEN INDEPENDENT PLUGIN IMPLEMENTATIONS.

Measure	**Result**
Both terminal / both graded	74 / 162
Claude-only / Codex-only terminal	11 / 3
Routing agreement	**236/250 (94.4%)**
Shared terminal-category agreement	**74/74 (100%)**
Joint-score Spearman	**0.8907**
Mean absolute score difference	6.85
Joint mean, Claude / Codex	76.27 / 78.23

TABLE VI

VALIDATION ARGUMENT: EACH CENTRAL DESIGN CLAIM IS SUPPORTED BY AN INDEPENDENT OBSERVABLE RESULT.

Claim	**Validation test**	**Observed result**	**Interpretation**
The funnel identifies outputs that are not usable summaries	Planted source-copy and off-topic populations	Both Plugins: 50/50 copies and 16/16 off-topic candidates terminated	The clearest eligibility failures are detected consistently without a runtime reference.
The evaluator does not reject strong summaries indiscriminately	Offline join against 58 reference reproductions	Both Plugins: 58/58 survived	Terminal gates preserve a known high-quality population.
The final ordering is safe for the five-candidate task	Compare the planted-bad candidate with intended survivors within every article	Both Plugins: 0/50 ranking inversions	Terminal tiers and graded scores place planted failures below useful candidates.
The design is reproducible across Agent hosts	Independent Claude and Codex implementations	94.4% routing agreement; 0.8907 score Spearman	The result follows the evaluation contract rather than one host-specific prompt trajectory.

IV. Limitations and Future Work

The present evidence covers 250 Japanese news summaries. The next experiment should apply the same Plugin-based funnel to larger datasets and to additional languages, testing whether routing, label calibration, and cross-host agreement remain consistent across broader data distributions and multilingual inputs.

V. Conclusion

The primary contribution is a reference-free evaluation architecture that separates eligibility from graded quality and assigns each method only the authority its evidence supports. Deterministic checks establish physical facts; embedding supplies mandatory assigned-article relevance evidence; isolated Agents judge central semantic validity and graded quality; independent Reviewers verify every consequential decision; and deterministic validators assemble ranked JSONL and reports. Delivering the design as two installable Plugins makes it executable and reproducible rather than prompt-dependent. The full 250-pair evaluation, 91.49% primary weak-label agreement, zero planted-bad inversions, 94.4% cross-host routing agreement, and 0.8907 score correlation jointly support the reliability of the resulting scores.